CSE - 641 Computer Vision

Weekly Project Report – 2

Ahmedabad University

Genotype-to-Phenotype Modeling of Alzheimer's Disease Using MRI Radiomics and Polygenic Risk Scores

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***Abstract*—Alzheimer disease is a disorder that affects the brain leading to loss of memory and progressive impairment of the thinking ability. Studies indicate that genetic factors as well as changes in the brain structure are significant in the pathogenesis of Alzheimer disease. However, the correlation between the inheritable genetic risk and the change in the brain structure is not well understood. The presented project seeks to examine the association between genetic vulnerability and brain structure by means of Polygenic Risk Scores (PRS) and MRI-based radiomic characteristics. The features of important brain areas including the hippocampus and the entorhinal cortex, will be extracted using MRI scans of the Alzheimer Disease Neuroimaging Initiative (ADNI) dataset. To determine inherited genetic risk, PRS will be calculated. Machine learning models will be then used to identify the relationship between genetic risk and features of MRI. The aim of the study is to gain more insight into the role of genetic factors in the early brain changes of the Alzheimer disease potentially before symptoms appear.**

I. Introduction

Alzheimer disease is an advanced disorder of the brain which gradually impairs the memory, ability to think and general brain performance. It is connected with when brain tissue is gradually lost throughout time. These changes are usually studied by structural magnetic resonance imaging (MRI), particularly in those parts of the brain that become affected at an early stage in the disease, the hippocampus, entorhinal cortex and temporal lobes [1]. Meanwhile, genome wide association studies of large sizes have enhanced the insights into the genetic etiology of Alzheimer. The genetic risk variants that have been identified through such studies are too many and predispose an individual with the probable cause of the disease [2]. These variants may be used together in a polygenic risk score (PRS) i.e., to sum up inherited genetic risk in a single value [3]. Both MRI markers and PRS can be used to understand the disease, but this is in two dimensions- brain structure and genetic risk.

Recently, scientists have begun using genetic and imaging data together in order to have greater insight into the influence of inherited risk on the brain prior to the emergence of symptoms. The goal of imaging genetics research is to investigate genetic risk or structure brain interactions during the early phases of the disease [4]. Moreover, radiomics enables the derivation of hundreds of shape and texture features of an MRI scan that are no longer measured in terms of volume [5]. With the convergence of PRS and radiomic features using machine learning models based on MRI, it is possible to investigate the effects of genetic risk on the brain structure. This will be a way of gaining a more clear evidence on how genetic factors lead to early structural changes in Alzheimer disease.

II. Dataset Description

The data used in this project is obtained from the **Alzheimer's Disease Neuroimaging Initiative (ADNI)**, a multi-centered longitudinal large-scale study established to identify and validate biomarkers for the early detection and monitoring of Alzheimer's disease progression.

The study participants are categorized into these primary groups:

- Cognitively Normal (CN)
- Significant Memory Concern (SMC)
- Early Mild Cognitive Impairment (EMCI)
- Late Mild Cognitive Impairment (LMCI)
- Alzheimer's Disease (AD)

ADNI provides multimodal data, including:

- Structural Magnetic Resonance Imaging (MRI)
- Positron Emission Tomography (PET) scans (FDG-PET and Amyloid PET)
- Cerebrospinal Fluid (CSF) biomarker values
- Genetic data (e.g., APOE genotype)
- Neuropsychological assessment scores (e.g., MMSE, CDR, ADAS-Cog)
- Demographic information

The data were collected across multiple phases based on the years it was collected, namely:

- ADNI-1
- ADNI-GO
- ADNI-2
- ADNI-3
- ADNI 4

These phases enable longitudinal analysis of disease progression over time. The dataset is cross-standardized across acquisition sites, ensuring consistency and reliability. Due to its comprehensive and longitudinal nature, the ADNI dataset is widely used in machine learning and deep learning research for classification, progression prediction, and biomarker discovery in Alzheimer's disease studies.

A. Dataset Selection

In this research, the entire data was then narrowed down to one subset to guarantee uniformity and radiomics analysis. Data contains structural MRI scans collected with the T1-weighted MPRAGE sequence, as part of the ADNI-2 and ADNI-3 phases owing to the high spatial resolution and standardized experimental procedures. Scans taken at 3 Tesla (3T) were used exclusively because the scans with high field strength offer better ratio of signal to noise and a better image as compared to lower field strength scans which are used in quantitative extraction of features. Baseline visits were chosen to ensure one scan per subject and avoid longitudinal bias.

TABLE I: Summary of Selected ADNI Dataset Parameters

Parameter	Selection	Description
ADNI Phases	ADNI-2, ADNI-3	These phases differentiate based on the number of years.
Imaging Modality	Structural MRI	Used for extracting structural brain features and radiomic biomarkers.
MRI Sequence	T1-weighted MPRAGE	Standard structural sequence widely used for brain morphology studies.
Field Strength	3 Tesla (3T)	Provides higher spatial resolution and better signal-to-noise ratio.
Visit Type	Baseline	Ensures each subject contributes a single scan, avoiding longitudinal duplication.
Subject Groups	CN, EMCI, LMCI, AD	Represents healthy controls, early impairment, and Alzheimer's disease patients.
Image Format	NIfTI (.nii)	Standard neuroimaging format compatible with most analysis pipelines.
Genetic Data	APOE Genotype	Known genetic risk factor strongly associated with Alzheimer's disease.

III. Literature Review

A. Longitudinal structural MRI-based deep learning and radiomics features for predicting Alzheimer's disease progression

The authors in this paper show how a longitudinal MRI-based paradigm can be utiliized to forecast the subjects who will progress to Alzheimer disease from mild cognitive impairment (MCI). They build a two-stage deep learning model on the T1-weighted images of the Alzheimer Disease Neuroimaging Initiative dataset. They first train a 3D ResNet18 model on individual visit MRI volumes to enable it to learn spatial brain features related to disease risk with a survival-based ranking loss. The learned feature representations are then used as input to a time-varying LSTM that trains to learn multiple MRI scans over time, accounting for irregular time intervals between visits. The attention mechanism identifies the most relevant time points. They also discard gray matter radiomics features like shape, intensity, and texture features, and discard them using PCA, and then combine them with deep features. The final model is being placed within a survival analysis framework to estimate individualized risk of time-to-conversion, rather than just binary classification.

B. Predicting Conversion from Mild Cognitive Impairment to Alzheimer's Disease Using a Vision Transformer and Hippocampal MRI Slices

The deep learning models are employed to predict the development of Mild Cognitive Impairment (MCI) to Alzheimer's Disease (AD) from MRI scans. Most of the existing studies were conducted using Convolutional Neural Networks (CNNs) because they possess the capability to learn spatial information from images of the brain. Some studies have been demonstrated to possess high accuracy, but there were problems such as data leakage, where data from the same patient was used for training and testing, which could potentially lead to biased results.

Recently, Vision Transformers (ViTs) have been proposed for image processing. Although Vision Transformers have been demonstrated to possess high accuracy for Alzheimer's classification, few studies have been conducted to explore their application for predicting MCI to AD conversion.

This paper continues from the existing studies by employing a Vision Transformer model that was pretrained on hippocampal MRI slices. It strictly separates data for each subject and compares the results to a 3D CNN model. The results indicate that Vision Transformers possess the capability to produce similar results to CNN models.

Polygenic Risk Scores provide an understanding of how combination of genes of a person may influence chance of developing Alzheimer's disease. The authors of the [7] explored many case control and by proxy studies data through a genome wide association method to determine genetic differences associated with Alzheimer's disease. The research shows that when PRS used in combination with APOE carrier status could stratify individuals on the risk and the median age at onset of high-risk individuals differs. This paper brings attention to the fact that it is better to combine the numerous genetic risk factors into one score that would further predict the disease susceptibility. Though the study mainly concentrates on the genetic side it also forms the basis of the association of genetic risk with alteration of the brain which show that early change in the structure of the brain could occur in people with high PRS. In the context of our project this provides the reasoning behind the integration of PRS and MRI-based radiomic features in the effort to investigate the development of cumulative genetic risk into subtle phenotypes in hippocampal and entorhinal cortex structures which would possibly support the earlier detection of the advancement of Alzheimer disease.

IV. Work Completed in Week-2

In Week 2, our group faced some difficulties while finalizing the parameters from the dataset. We also went through some papers to gain a clear understanding of this domain.

- In Week-2, our problem identification is clear, which is to classify patients based on CN, MCI, and AD, and the data type to be used for classification, which could be MRI, PET, and so on.

- We explored the Alzheimer's disease data set, and the data type and data format were clear to us.
- Further research papers and literature were explored to find possible improvements and better approaches for our project.

In conclusion, the week was spent on developing the clear vision of the dataset.

V. Plan for Week-3

During Week-3, we will download and merge our dataset, and start working on the implementation of the required end-to-end pipeline. The tasks to be done are as follows:

- By the close of Week-3, the necessary data of ADNI dataset will be downloaded and organized accordingly.
- The python environment having all the required libraries will be successfully configured.
- MRI images will be loaded and checked in terms of right format and structure.
- Genetic data will be ready and compared to the respective MRI subjects.
- An elementary processing pipeline of feature extracting and PRS integrating will be developed.
- A short trial on a small group of subjects will be done to test the correct functionality of the pipeline.

References

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